PROJECT REPORT

SUBMITTED BY – TUSHAR VIDHURI

COURSE- AWS INTERNSHIP BATCH 1.1

COLLEGE – IIMT ENGINEERING COLLEGE , MEERUT

INTERNSHIP DURATION- 28/SEP/26 -28/OCT/26

ORGANISGATION NAME – EWB COURSES

Project Title –

AWS VPC Peering: Establishing and Testing Private Network Connectivity Using Amazon EC2
Organization: EWB Cloud Labs
Project Domain: Cloud Computing and AWS Networking
Platform: Amazon Web Services (AWS)
Project Type: Practical Cloud Networking Lab
Region: Mumbai (ap-south-1), or the Region actually used
Technologies: Amazon VPC, Amazon EC2, VPC Peering, Route Tables, Security Groups, Internet Gateway, SSH, and ICMP

Introduction –

Amazon Web Services (AWS) provides cloud computing services that allow users to create and manage virtual networks, servers, storage, and other infrastructure resources.

In this project, we implement private network connectivity between two separate Amazon Virtual Private Clouds (VPCs). We deploy two private application servers in different VPCs and use a third EC2 instance as a public bastion host for secure administration.

The two application servers must communicate with each other through their private IPv4 addresses without assigning public IP addresses to them. To achieve this, we establish a VPC Peering connection, configure appropriate route tables, and apply security group rules that permit only the required traffic.

We first perform a baseline ping test before establishing peering to demonstrate that connectivity is unavailable. After configuring the peering connection and routes, we test communication in both directions. We also verify SSH access to the private servers through the bastion host and perform a controlled failure test by temporarily removing an ICMP security rule.

This project provides practical experience in AWS network architecture, private connectivity, access control, and systematic troubleshooting.

What Is Our Requirement?

The main requirement of this project is to establish a secure private network connection between two VPCs and demonstrate how routing and security configurations affect network connectivity.
To complete the project, we must:

1. Create two separate VPCs with non-overlapping IPv4 CIDR blocks.
2. Create one public subnet and one private subnet in VPC A.
3. Create one private subnet in VPC B.
4. Launch three EC2 instances: one public bastion host and two private application servers.
5. Assign the required private IP addresses to the instances.
6. Create and configure an Internet Gateway for VPC A.
7. Create three custom route tables and associate them with the correct subnets.
8. Configure security groups to allow only the required SSH and ICMP traffic.
9. Establish and activate a VPC Peering connection between VPC A and VPC B.
10. Configure forward and return routes through the peering connection.
11. Demonstrate a failed ping before peering and successful ping tests after configuration.
12. Verify private SSH access to the application servers through the bastion host.
13. Perform a controlled ICMP failure test and restore connectivity.
14. Capture screenshots of the actual configuration and test results.
15. Clean up the AWS resources after completing the project.

The final objective is to prove that two private application servers in separate VPCs can communicate through private IP addresses while remaining inaccessible through direct public IP access.

Project Architecture –

The project architecture consists of two VPCs, three EC2 instances, three custom route tables, one Internet Gateway, three security groups, and one VPC Peering connection.
VPC A — 10.10.0.0/16
VPC A contains two subnets:

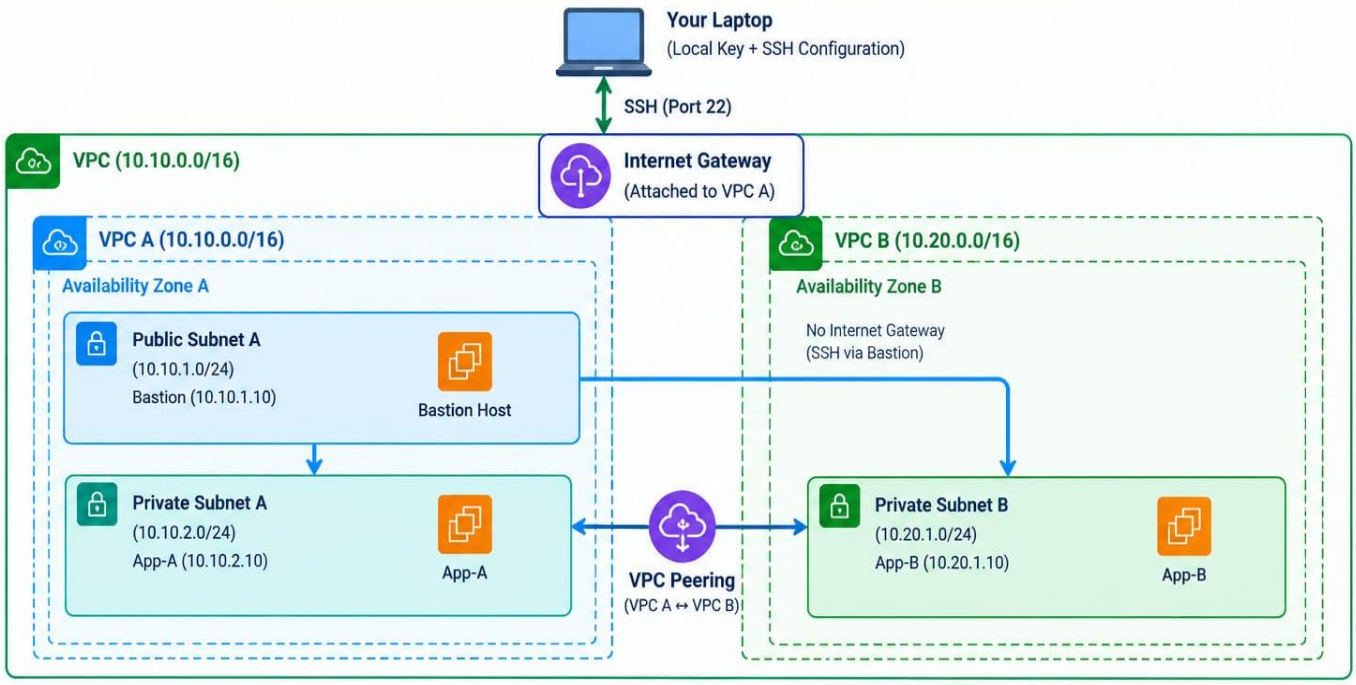
- Public Subnet A: CIDR 10.10.1.0/24. It contains the bastion host with private IP 10.10.1.10 and a public IPv4 address.
- Private Subnet A: CIDR 10.10.2.0/24. It contains App-A with private IP 10.10.2.10 and no public IP address.

VPC A is connected to an Internet Gateway to provide internet connectivity to the bastion host.
VPC B — 10.20.0.0/16
VPC B contains one private subnet:

- Private Subnet B: CIDR 10.20.1.0/24. It contains App-B with private IP 10.20.1.10 and no public IP address.

VPC B does not have an Internet Gateway. Its private workload communicates with VPC A through the VPC Peering connection.
VPC Peering Connection
The VPC Peering connection provides a private network path between VPC A and VPC B. The route tables on both sides direct traffic to the peer VPC through this connection.
Administration and Application Traffic
Administrative access follows this path:
Laptop → Bastion Host → App-A or App-B
Application connectivity follows this path:
App-A ↔ VPC Peering Connection ↔ App-B
The bastion host is used for SSH administration. The application servers communicate with each other using private IPv4 addresses.
VPC Peering does not provide VPC B with internet access through VPC A's Internet Gateway.

AWS Network Architecture – VPC with Peering Connection



Key Components

- VPC A: 10.10.0.0/16 (with Internet Gateway)
- VPC B: 10.20.0.0/16 (no Internet Gateway)
- Public Subnet A: 10.10.1.0/24 (Bastion Host)
- Private Subnet A: 10.10.2.0/24 (App-A)
- Private Subnet B: 10.20.1.0/24 (App-B)
- VPC Peering: Connects VPC A and VPC B

Traffic Flow

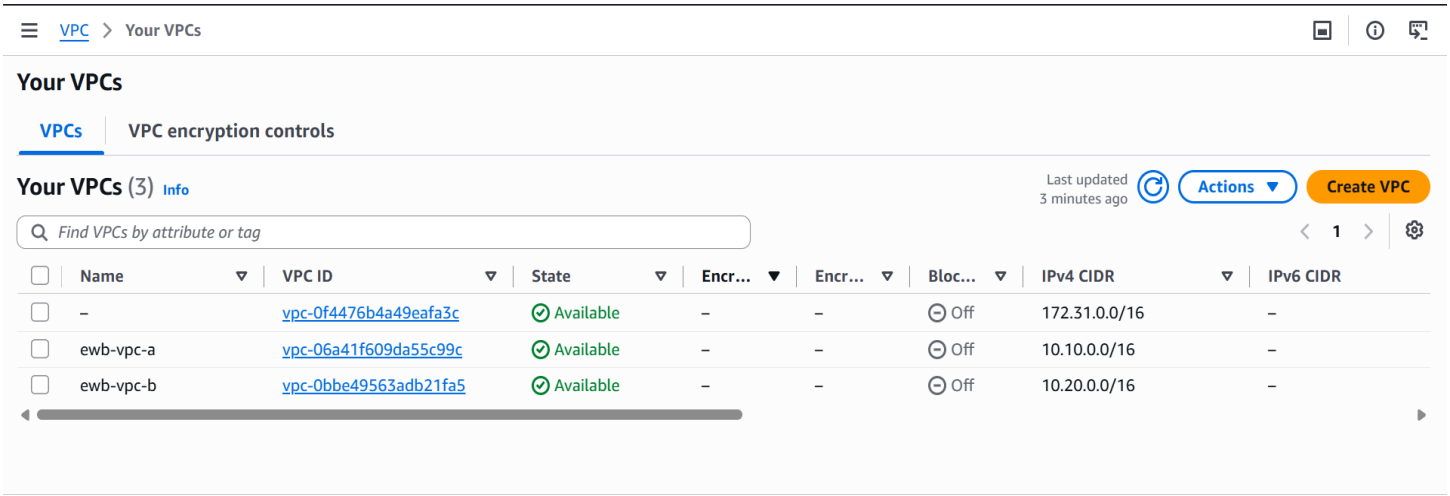
1. Your laptop connects to Bastion Host via SSH (Port 22)
2. Bastion Host (in Public Subnet A) connects to App-A (in Private Subnet A)
3. App-A and App-B communicate through VPC Peering (Private IPs)

IP Address Summary

- VPC A CIDR: 10.10.0.0/16
- Public Subnet A: 10.10.1.0/24 (Bastion: 10.10.1.10)
- Private Subnet A: 10.10.2.0/24 (App-A: 10.10.2.10)
- VPC B CIDR: 10.20.0.0/16
- Private Subnet B: 10.20.1.0/24 (App-B: 10.20.1.10)

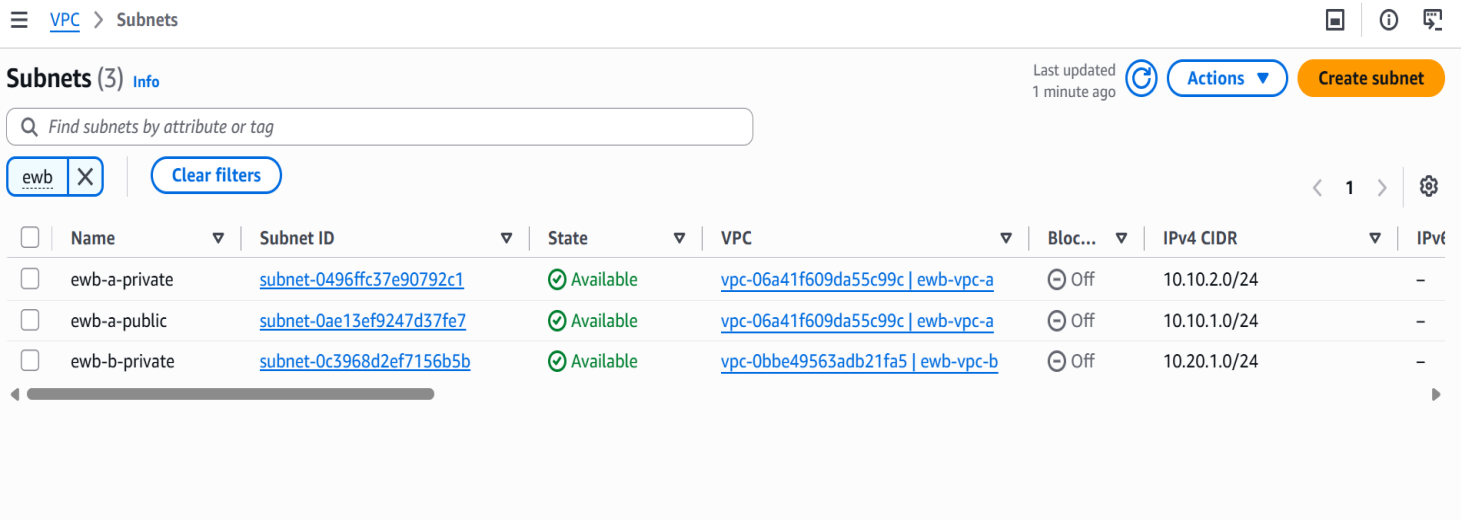
VPC Creation –

Two VPCs were created: ewb-vpc-a and ewb-vpc-b. VPC A uses CIDR 10.10.0.0/16, while VPC B uses 10.20.0.0/16. These VPCs provide separate networks for private communication



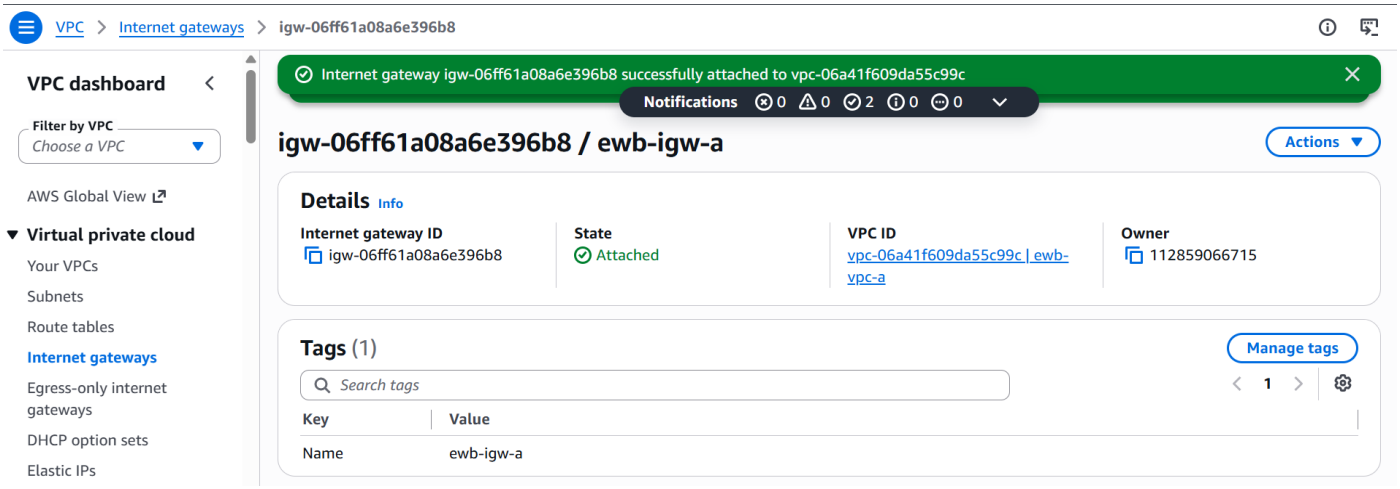
Subnets Creation –

Public and private subnets were created in VPC A, while a private subnet was created in VPC B. The subnets organize EC2 instances and provide separate network environments.



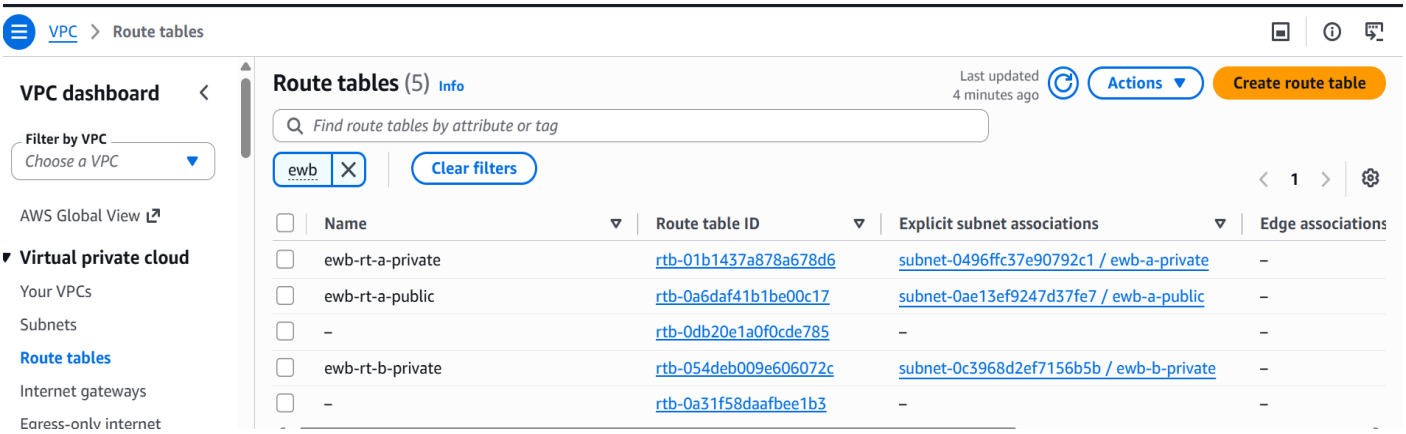
Internet Gateway –

An Internet Gateway was attached to VPC A to provide internet connectivity to resources with appropriate public IP addresses and routes. VPC B does not use an Internet Gateway.

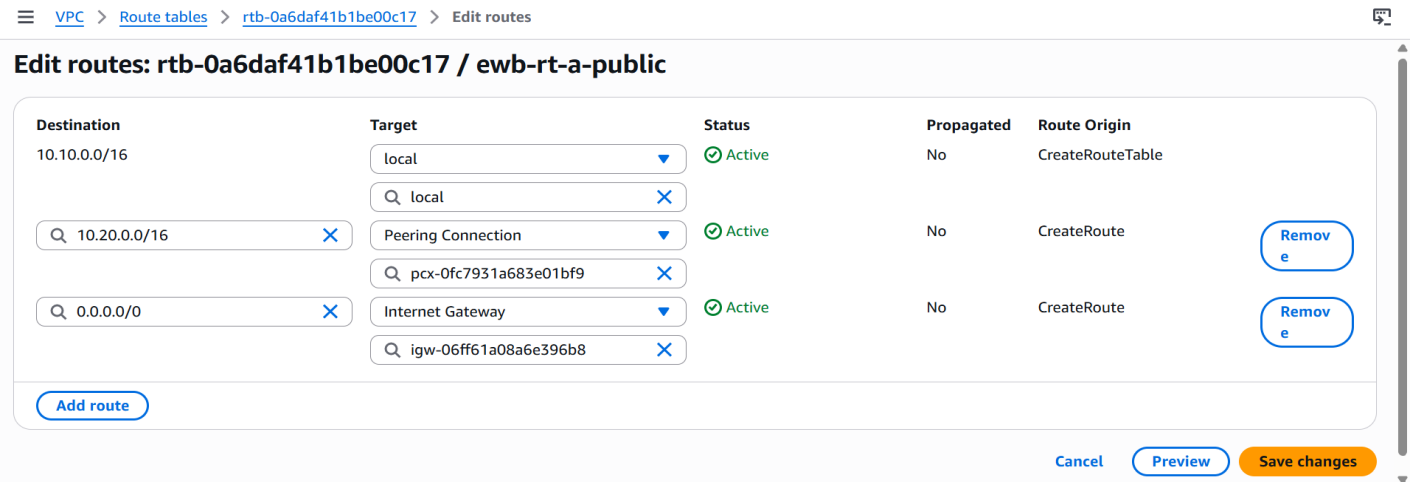


Route tables with subnet Association –

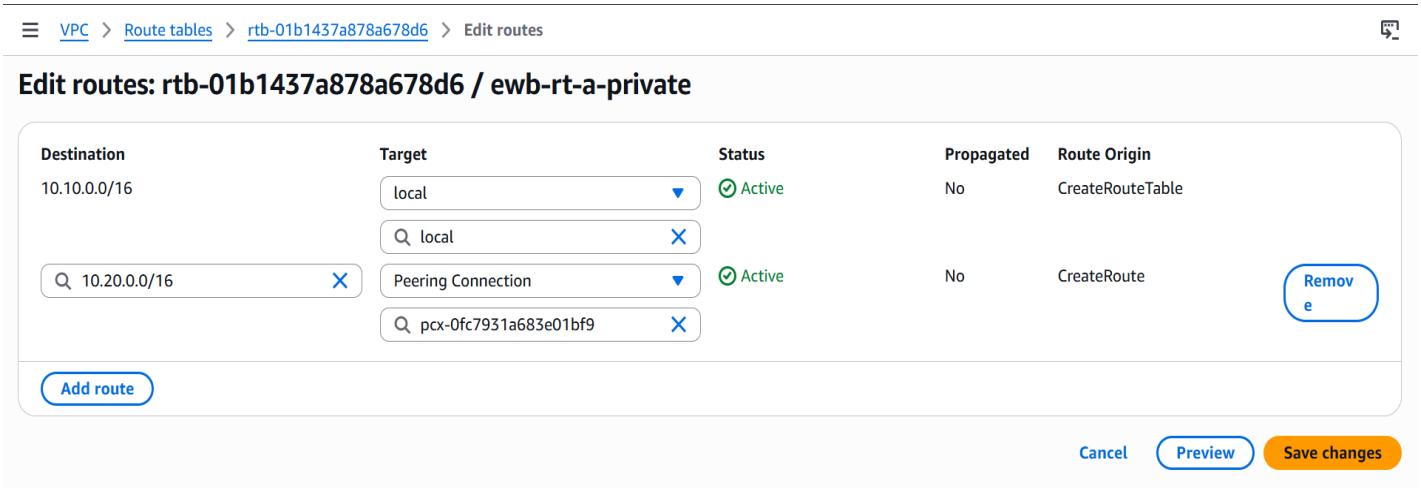
Routes were added to the relevant route tables. VPC A received a route to 10.20.0.0/16, and VPC B received a route to 10.10.0.0/16, both through the VPC Peering connection.



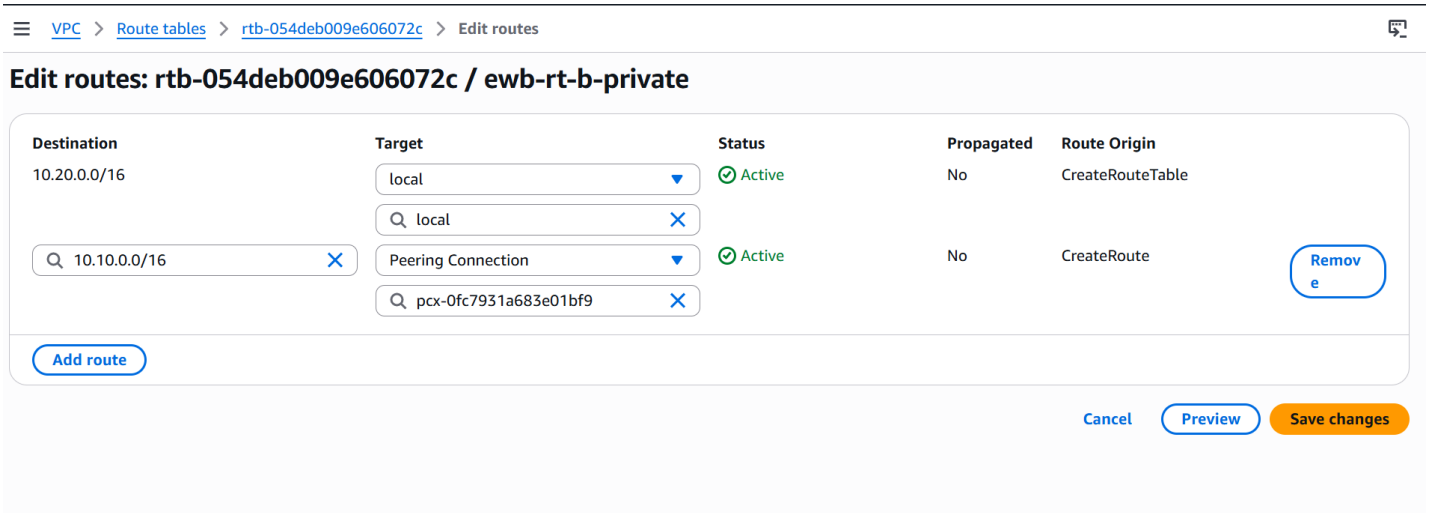
Route Table –Vpc A public –



Route table – Vpc A private –

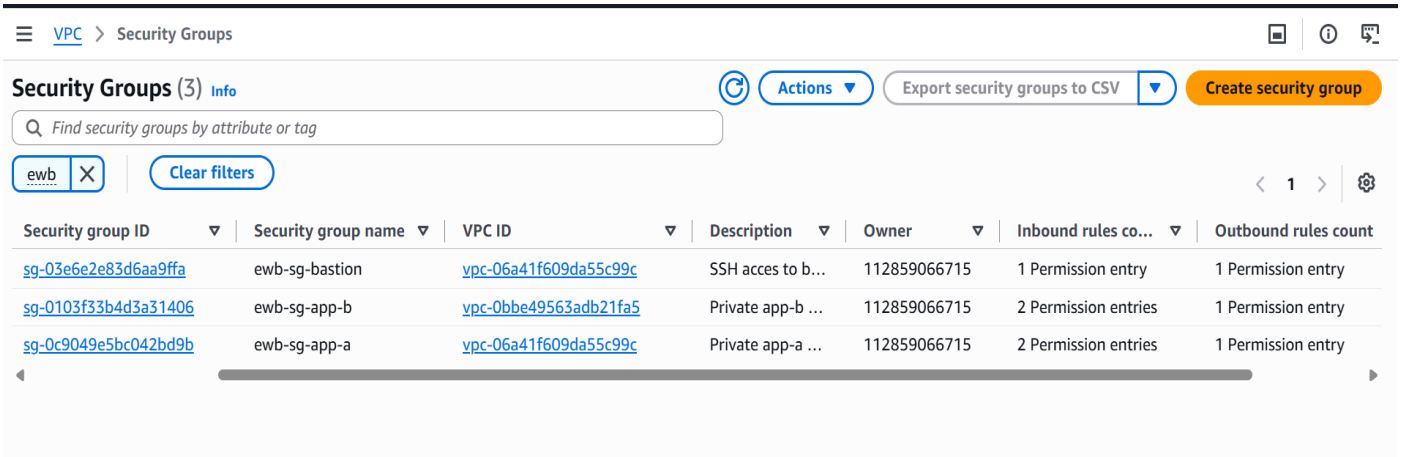


Route Table – Vpc B private –

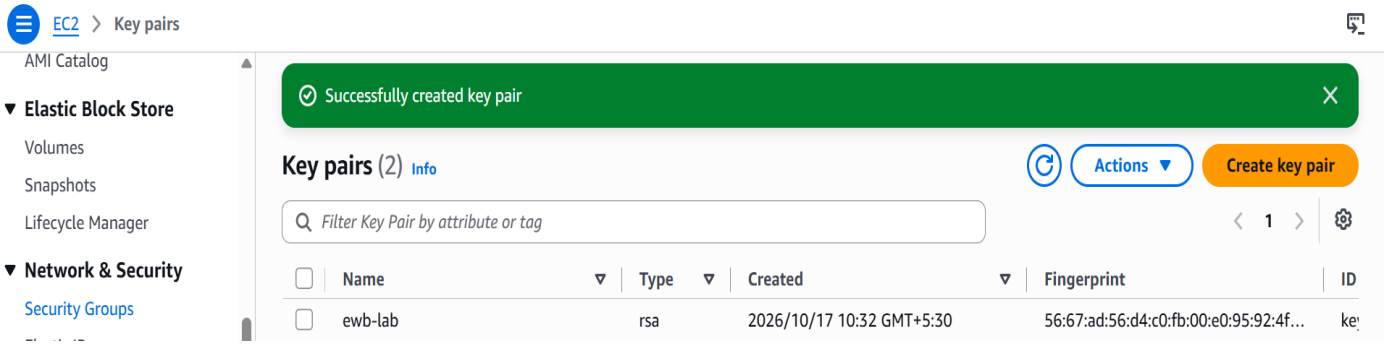


Security Group –

Security group rules were configured to allow SSH access from the required sources and ICMP traffic between App-A and App-B. These rules control which traffic is permitted.

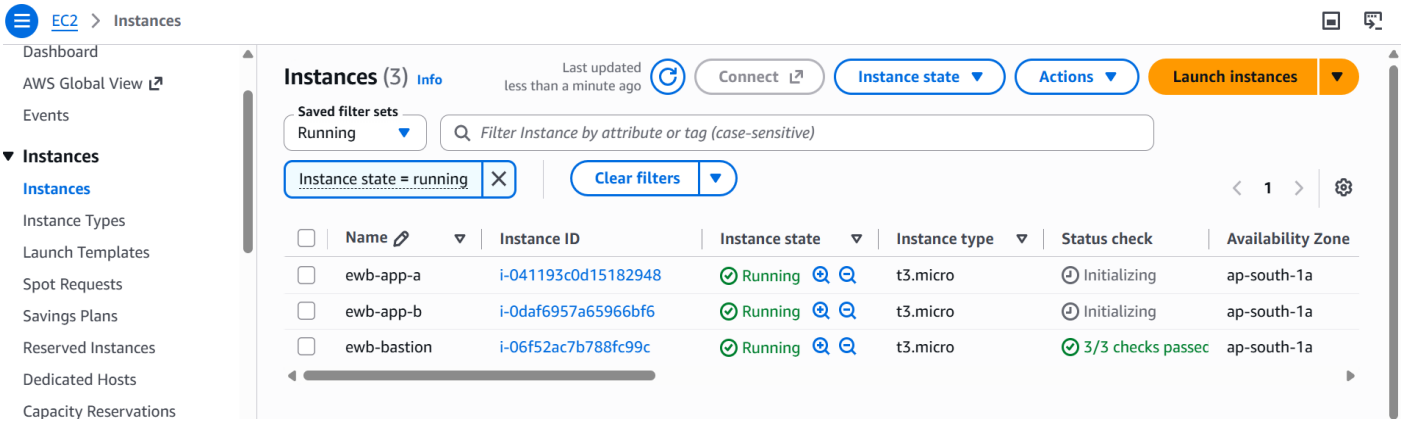


Key pair –



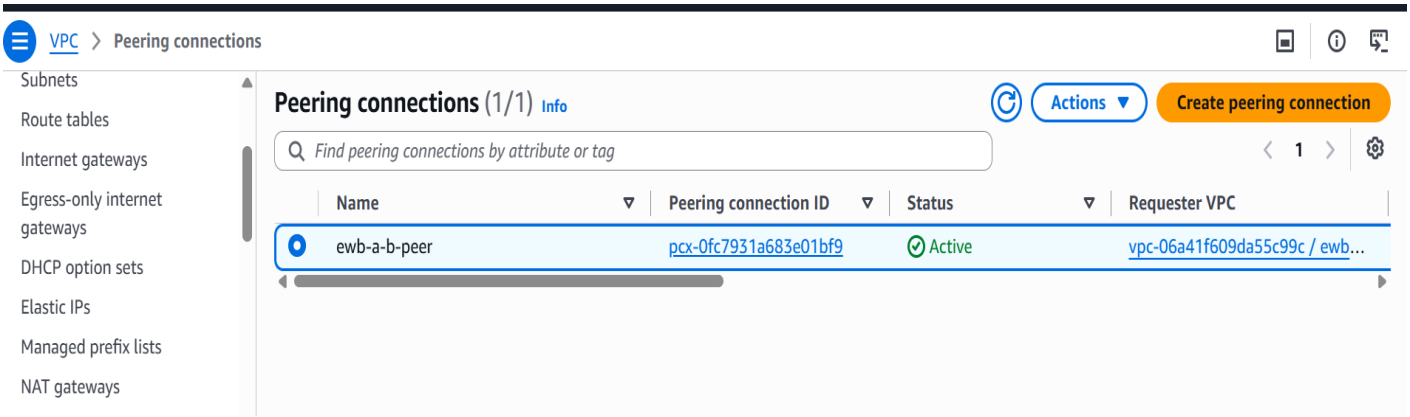
EC2 Instances Creation –

Three EC2 instances were configured: a Bastion Host, App-A, and App-B. App-A and App-B use private IP addresses and do not require public IP addresses for VPC-to-VPC communication.



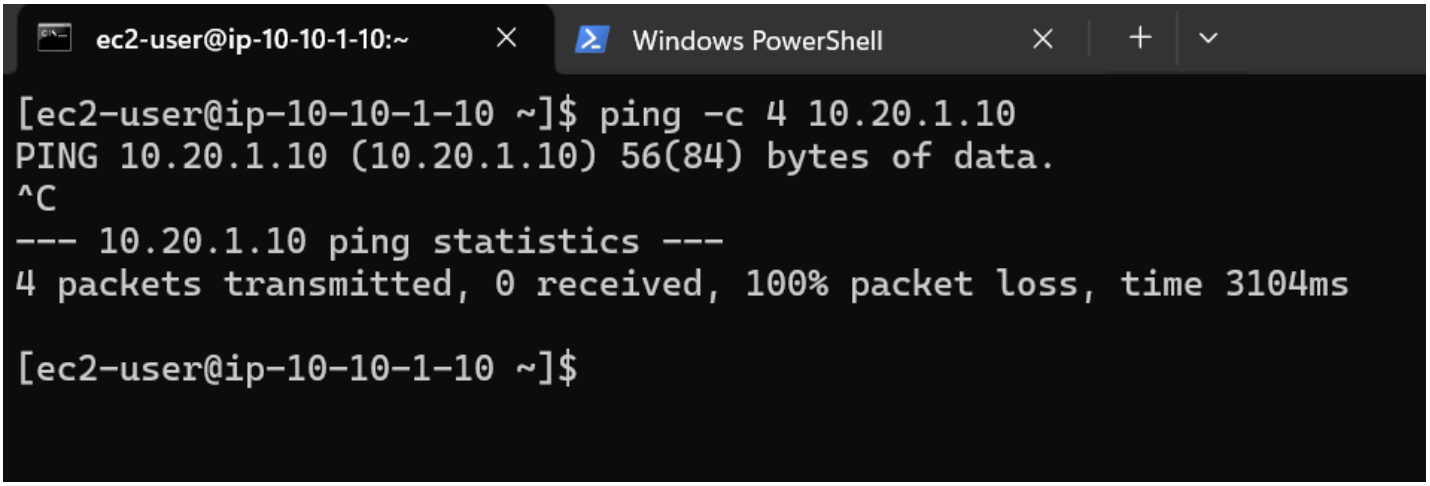
Peering Connection –

A VPC Peering connection named ewb-a-b-peer was created between VPC A and VPC B. It provides a private network connection between the two VPCs.

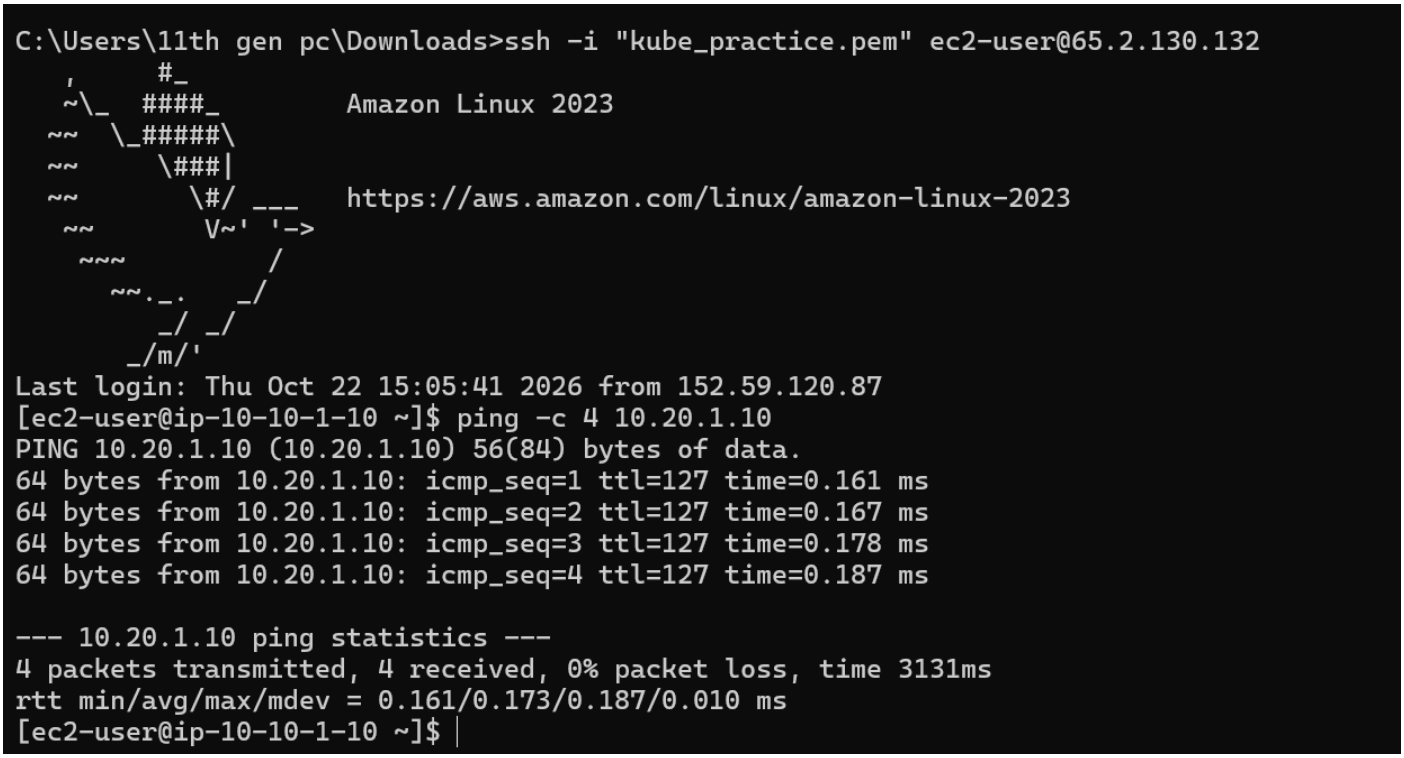


Base Test case –

Before configuring VPC Peering and the required routes, a ping test from App-A to App-B failed. This demonstrated that private connectivity was not yet established.



Bastion to app b –



Ping status from A to B –

After configuring the peering connection, routes, and security group rules, App-A was able to communicate with App-B using its private IP address 10.20.1.10. The reverse ping test was also performed from App-B to App-A

```
[ec2-user@ip-10-10-1-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.173 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.205 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.181 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.197 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3108ms
rtt min/avg/max/mdev = 0.173/0.189/0.205/0.012 ms
[ec2-user@ip-10-10-1-10 ~]$
```

Ping status from B to A –

```
[ec2-user@ip-10-10-1-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.173 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.205 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.181 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.197 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3108ms
rtt min/avg/max/mdev = 0.173/0.189/0.205/0.012 ms
[ec2-user@ip-10-10-1-10 ~]$
```

Conclusion –

This project successfully demonstrates the configuration and testing of private connectivity between two AWS VPCs using VPC Peering. By configuring route tables and security groups, we can enable communication between private EC2 instances. The controlled failure test further demonstrates how security rules affect network connectivity. This project provides practical knowledge of AWS networking and troubleshooting.

